Valuing and Pricing Financial Datasets

From Predictive Power to Economic Value

Research Question

How should a financial dataset be priced?

- Data is the input that forms investor beliefs; beliefs determine portfolios; portfolios aggregate into order flow; order flow determines prices. *The distribution of data access directly determines how efficiently prices aggregate information across the economy.*
- As quantitative methods diffuse broadly, the scarce factor shifts from modeling skill to proprietary data — whoever controls the signal controls the edge.
- Yet data markets are almost entirely **private and opaque**: no observable prices, no standard contracts, no theory of how they work.

Two questions this paper addresses:

1. What is a dataset worth to a single investor?
2. How does a data vendor optimally price and license that dataset?

Value = PV of Incremental Trading Profits

Pay for data what the extra profits it generates are worth, discounted over your investment horizon:

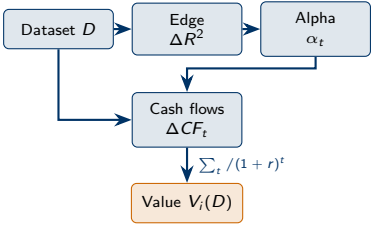
$$V_i(D) = \sum_{t=1}^{T} \frac{\mathbb{E}[\Delta CF_{i,t}(D)]}{(1+r_i)^t} - C_i(D)$$

$\Delta CF_{i,t}$ = incremental trading profit from D in period t; $C_i(D)$ = acquisition cost.

$\Delta CF_{i,t}$ **is the product of two objects:**

- the *predictive edge* D adds over existing information (ΔR^2)
- the *capital profitably deployable* on that edge

Both are endogenous to market structure. Three dynamics determine them.



Value is investor-specific ($\bar{w}_i$, γ_i, style), forward-looking, and endogenous to market structure.

Three Dynamics That Complicate Valuation

Each component of $\Delta CF_{i,t}(D)$ is endogenous:

1. Market Size

Even with unlimited capital, cash flows from a signal are **bounded by market depth**. A signal on a small or illiquid universe hits a hard ceiling on extractable profit regardless of predictive accuracy.

2. Deployable Capital

Value scales with AUM **up to the point where position size exceeds market absorption capacity**. Beyond that, price impact erases gains. The AUM–value relationship is concave, not linear.

3. Adoption

As n investors trade the same signal, they **collectively reveal it to the market**. Cash flows shrink for all holders as the mispricing is corrected faster. Value is endogenous to who else owns the data.

Together these dynamics explain why *statistical accuracy is neither necessary nor sufficient for economic value* — the central puzzle this paper resolves.

Contribution

Farboodi, Singal, Veldkamp, and Venkateswaran (2021) derive the private value of a dataset to a single investor and explicitly set aside two next steps. This paper takes both, plus a third.

1. **Prediction-to-precision bridge.** Maps observable metrics (ΔR^2, MSE) to the signal precision parameters that enter the valuation formula — making the cash-flow framework computable from numbers practitioners actually report.

2. **Equilibrium transaction prices.** Derives the vendor's optimal licensing problem: how many licenses to sell and at what price. Private value $\neq$ transaction price.

3. **Competitive adoption as a first-order mechanism.** As n investors share and trade the same signal, crowding erodes $\Delta CF_{i,t}$ for all holders. Introduces *dataset capacity* $C(D)$ — the rate at which this erosion occurs — as a formal object.

Literature

Five strands — one gap

Strand	Key insight	Role here
Costly information	Grossman-Stiglitz (1980): informed investors must earn rents or no one gets informed	Why datasets have value at all
Data valuation	Farboodi et al. (2021): private value via sufficient statistics	Direct foundation; extended here
Info. monopoly	Admati-Pfleiderer (1986): seller faces price vs. adoption tradeoff	Vendor pricing framework
Crowding & decay	Kyle (1985); Holden-Subrahmanyam (1992); McLean-Pontiff (2016)	Micro-foundation for erosion
Club goods	Buchanan (1965): optimal club size under congestion	Structure of the n^* problem

The gap: No existing paper derives a transaction price for data accounting for both investor heterogeneity and endogenous value erosion.

Model

Setup: N risky assets; CARA investors heterogeneous in AUM $\bar{w}_i$, risk aversion γ_i, and style $\mathcal{Q}_i$. Dataset D produces a Gaussian signal about dividend innovations. Noisy rational expectations equilibrium.

Three components:

1. **Prediction-precision bridge** — connects observable ΔR^2 to signal precision $K_i(D)$:

$$K_i(D) = \frac{\Delta R^2(D) \cdot \sigma_y^2}{\mathrm{MSE}_D \cdot \left(1 - \Delta R^2(D)\right)}$$

2. **Private value** (closed form, proportional to AUM, convex in ΔR^2):

$$V_i(D) \approx \frac{\bar{w}_i}{2\gamma_i} \cdot \frac{\Delta R^2(D)}{1 - \Delta R^2(D)} \cdot \frac{\sigma_y^2}{\mathrm{MSE}_D} - C_i(D)$$

3. **Crowding** — n investors sharing the signal trade more aggressively (Holden-Subrahmanyam 1992), raising Kyle $\lambda(n)$ and reducing each investor's profit. $C(D)$ parametrizes how fast aggregate informed order flow becomes detectable in the market.

Key Results

Proposition 1 — Optimal Licensing

Vendor maximizes revenue at $n^* = (C(D) + 1)/2$ licenses. Low-capacity datasets should be licensed exclusively; high-capacity datasets sustain broad licensing.

Proposition 2 — First-Mover Value Premium

The k-th adopter captures more value than the $(k + 1)$-th. Premium is largest at the exclusive-to-shared transition and declines monotonically. Early movers capture peak alpha; late movers may find the data worthless.

Proposition 3 — Information Seller's Paradox

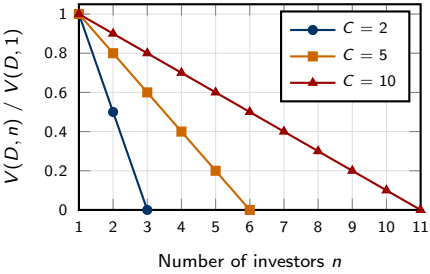
High accuracy ($\Delta R^2 \approx 1$) can coexist with zero or negative economic value when capacity is low and adoption is broad.

Further: Large investors are more crowding-sensitive (Prop. 4); revenue is monotone in quality *and* capacity (Prop. 6).

Visual Intuition

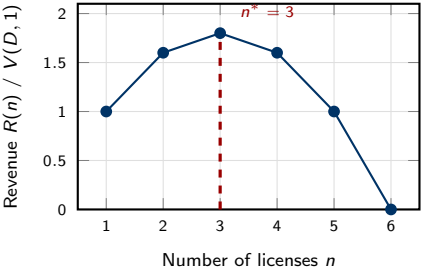
Inverted Network Effect

Per-investor value as adoption grows (Props. 2 & 3)



Vendor's Optimal Licensing

Revenue $R(n) = n \cdot V(D, n)$, normalized; $C = 5$



Next Steps

Immediate

- Derive $C(D)$ from Kyle/Holden-Subrahmanyam primitives — replace the imposed decay form with a derived object
- Solve the heterogeneous-investor vendor problem fully
- Dynamic adoption: alpha lifecycle as n grows endogenously

Extensions

- *Algorithmic trading.* AI traders implicitly collude on the cooperative trading pace (Goldstein and Dou) — n^* is increasing in how algorithmic the buyer pool is
- *Empirical implications.* Comomentum, 13F overlap, and post-publication alpha decay as observable proxies

Thesis goal: A complete theory of data market equilibrium — private value, transaction price, adoption dynamics, and vendor strategy.